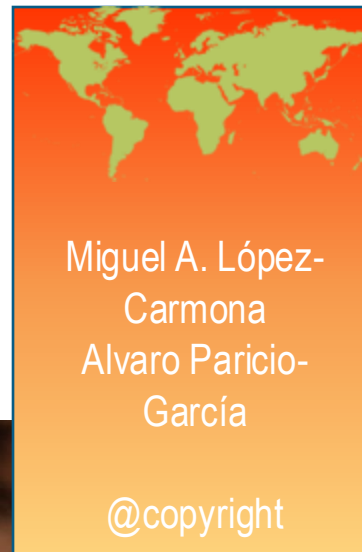


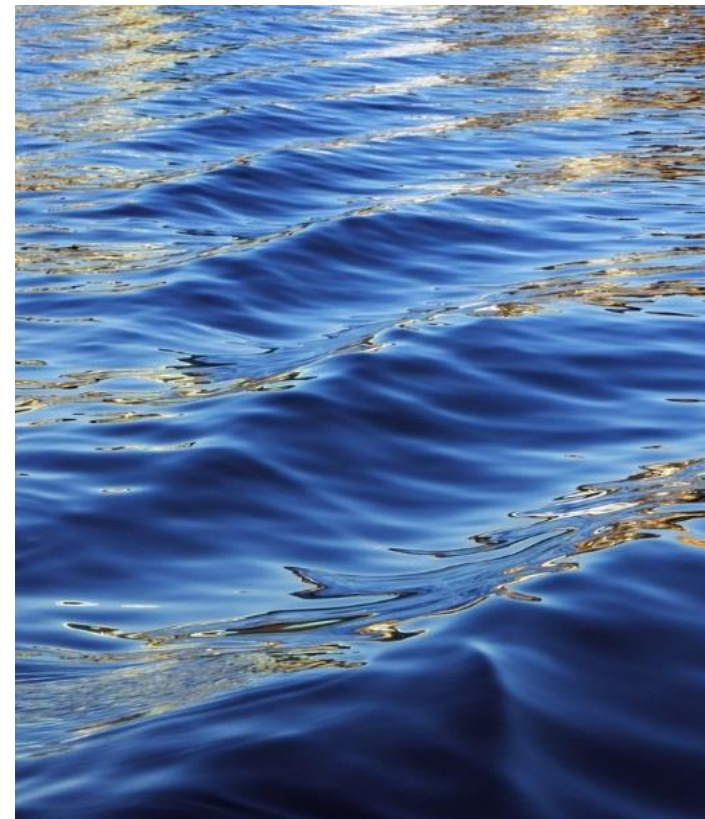


Push-pull and Segmentation



Miguel A. López-
Carmona
Alvaro Paricio-
García

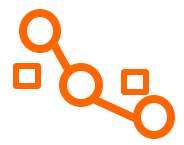
@copyright



Co-funded by the
European Union

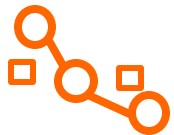


Universidad
de Alcalá

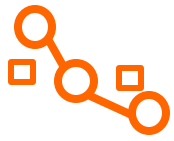


Learning objectives

- Push vs Pull Systems
- Segmentation Strategies
 - Products
 - Supply chains
- Uncertainty management



Push vs Pull Processes



You can learn almost anything about
logistics from a sandwich store.

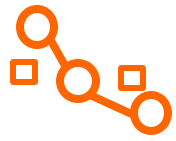
How many different sandwiches can be made?



Sandwich = Bread (3) + Protein (5) + Spread (4) +
Topping (5) --> 3*5*4*5 = 300 types of sandwich

- 1) Make to Order (MTO) Solo cuando lo piden
- 2) Make to Stock (MTS) Tener ya preparado lo usual
- 3) Engineer to Order (ETO) Crear algo nuevo





Pull vs Push Processes

■ Push

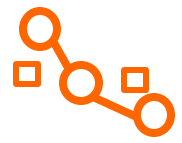
- Execution is performed in anticipation of an order.
- Demand is forecast
- Proactive process based on projected need/demand

■ Pull

- Execution is performed in response to a command
- The demand is real and is known with certainty.
- Reactive process based on actual need/demand

■ Limit between Pull and Push

- Point at which push processes are separated from pull processes



The story of three sandwiches



Ready turkey wrap



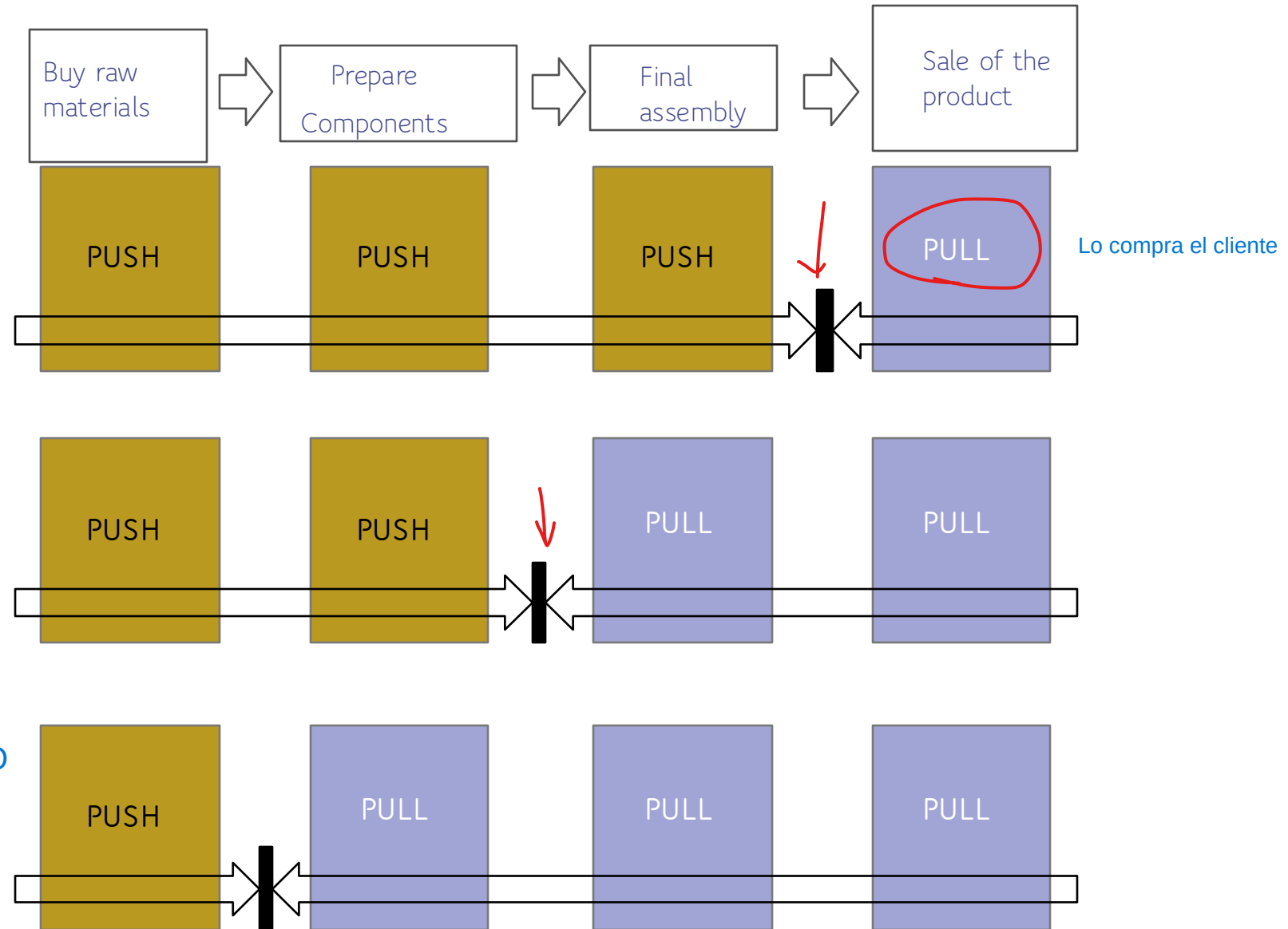
Gourmet Sandwich

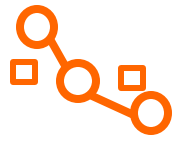


Design

MTO

ETO

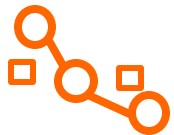




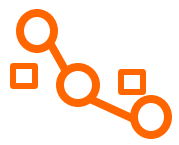
Push and Pull processes

- Mixed systems are common: Where is the Push-Pull point?
 - Promote raw and undifferentiated products or components.
 - Pulling the finished product
- Benefits of mixed systems
 - Enables efficient mass customization (Postponement)
 - Allows grouping of products: aggregate demand
- Fundamental principles
 - Maximizes external variety with minimal internal variety
 - Keep in-process inventory as raw as possible (RAP).

RAP



Segmentation



SC segmentation

In fact . . .

Companies operate multiple supply chains.

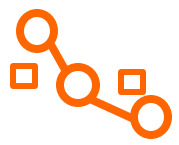
There is no one-size-fits-all supply chain

Companies segment to match the right method with the right product/customer/supplier combination

Companies can segment products, customers, suppliers, etc.

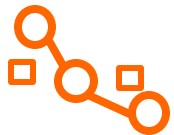
Segmentation only makes sense if you do something different in the way you buy, manufacture, move, store or sell!

- Purchasing and sourcing
- Demand forecasting/planning
- Inventory planning
- Inventory control
- Storage / Material Handling
- Order management
- Transportation Management
- Customer Service

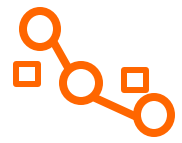


SC segmentation

- How many segments (basic rules)
 - **Homogeneity** within the segment
 - **Heterogeneity** among different segments
 - **Critical mass** - large enough to be worthwhile
 - **Pragmatics** - useful and communicable dimensions
- How can I segment my customers or suppliers?
 - Lead time (delivery time) Service level
 - Purchase history Volume or size of orders
 - Geography Demography
 - Sales trend Channel segmentation
 - Strategic importance
- How can I segment my products?
 - Physical characteristics (value, size, density, etc.)
 - Demand characteristics (sales volume, volatility, sales duration, etc.)
 - Supply characteristics (availability, location, reliability, etc.)



Distribution of SKUs (Stock Keeping Units)



Product segmentation

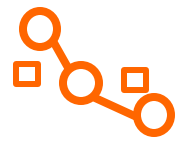
- Local grocery store
 - ~20,000 SKUs
 - Categories: Canned Food, Frozen & Perishable Food, Perishables
- Dry goods analysis (~8,000 SKUs)
 - 1,156 M SKUs sold in 1 year
 - Number of units sold per SKU
 - Half 144
 - Median 72
 - Fashion 0
 - Standard deviation 0
- Higher sales?
- Highest sales day?



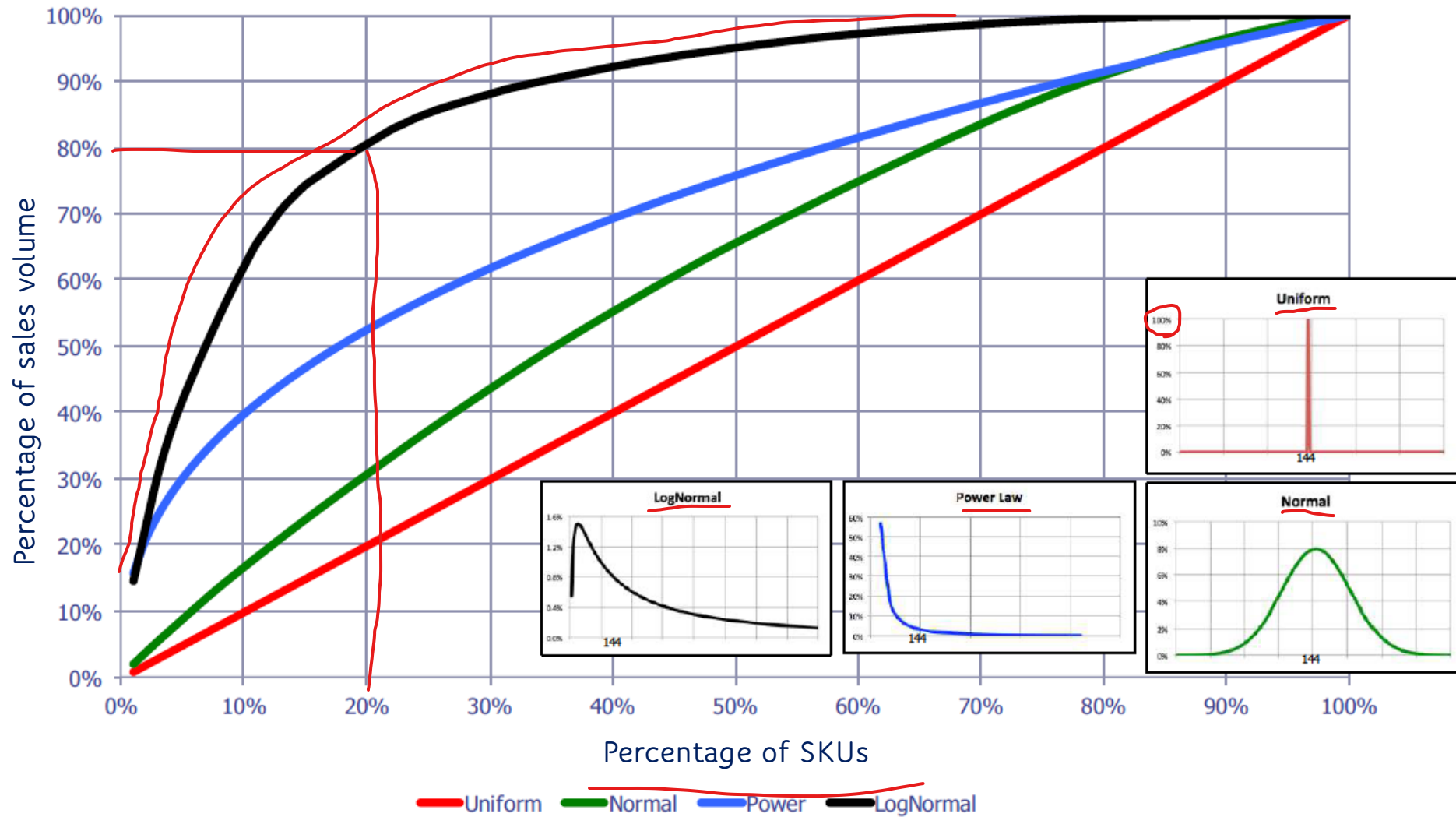
More sold

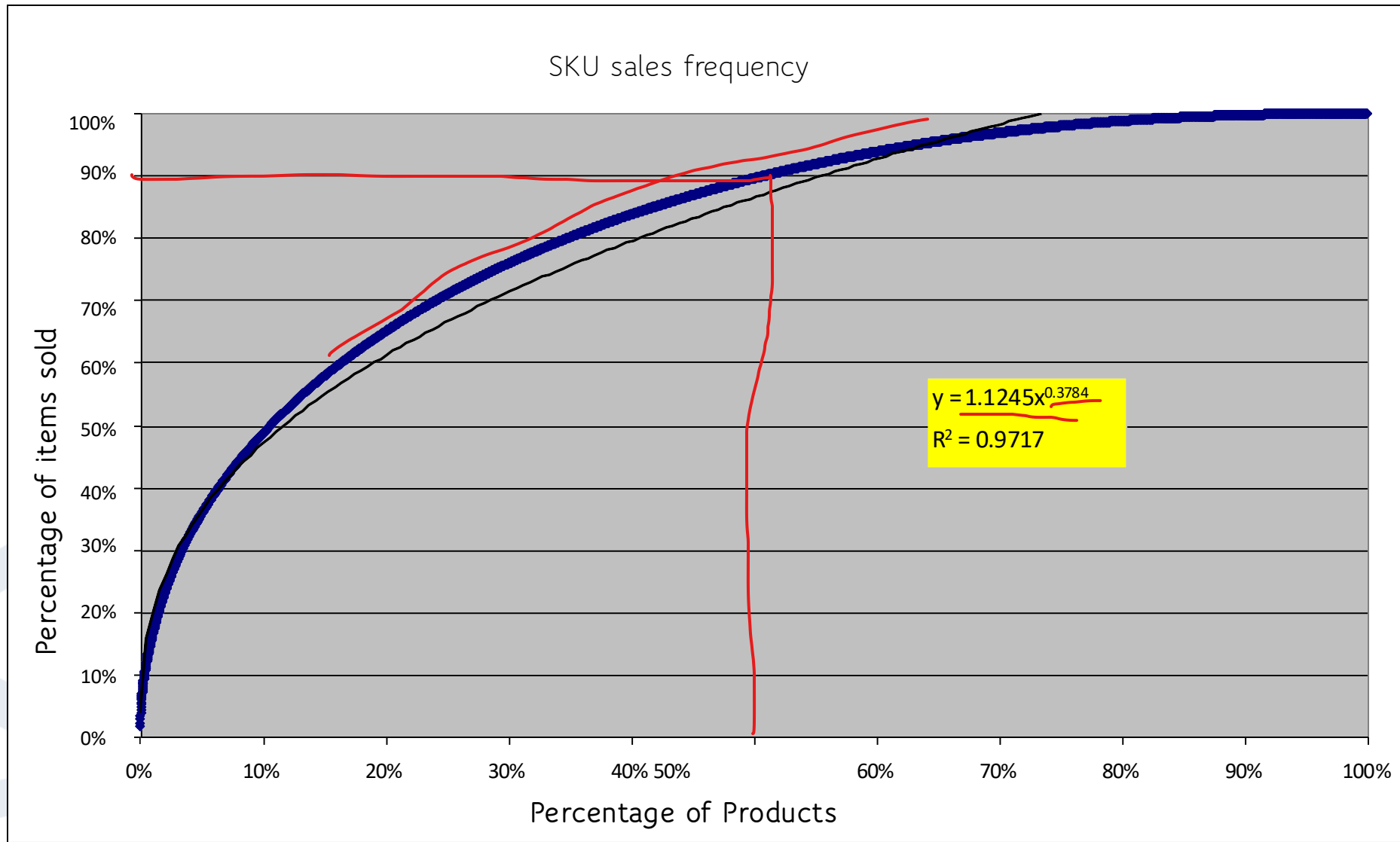
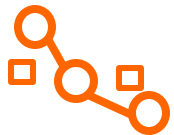
1. MILK 1.5l La Asturiana
2. Toilet paper
3. Fontvella BOTTLED WATER
4. Pipas Perez
5. Tuna Calvo

How are the products distributed in terms of sales volume: Uniform, Normal, Other?

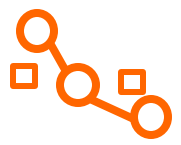


Possible product distributions





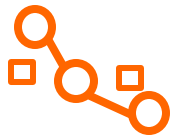
Example of the Power Law, $y = ax^k$



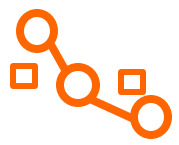
Power law

- Exceptionally common in social and physical systems
 - Severity of hurricanes and earthquakes
 - Income of a population (Pareto's Law)
 - Website visits (Nilsen's Law)
 - Frequency of words in any language (Zipf Law)
 - Frequency of digits in tables (Benford's Law)
 - Frequency of citations in literature (Lotka's Law)
 - Metabolic rates of animals relative to their mass (Kleiber Law)
 - Customer and product profitability
 - Traffic volume distribution in lanes
 - Student issues in a classroom

Conclusion: The distribution of many phenomena around a single population follow a Power Law relationship



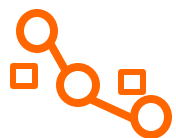
ABC Analysis



Segmentation: ABC Analysis

- Class A items-the few important ones
 - Very few high impact articles are included.
 - They require the utmost attention and management.
 - Many exceptions are expected to be made.
- Class B items-the middle part
 - Many elements of moderate impact (sometimes most).
 - Automated control with management by exception.
 - Rules can be used for A (but there are usually too many exceptions).
- Class C items-the many trivial ones
 - Many, if not most, of the elements that generate minor impact
 - Control systems should be as simple as possible.
 - Reduce wasted management time.
 - Grouping common regions, suppliers and end users.

These are arbitrary classifications!



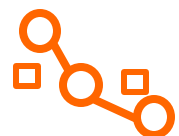
Segmentation: ABC Analysis

Part ID	Price	Annual Demand	Annual Value
1	2,25 €	260	585,00 €
2	2,85 €	43	122,55 €
3	1,50 €	21	31,50 €
4	0,77 €	388	298,76 €
5	4,45 €	612	2.723,40 €
6	6,10 €	220	1.342,00 €
7	3,10 €	110	341,00 €
8	1,32 €	786	1.037,52 €
9	12,80 €	14	179,20 €
10	24,99 €	334	8.346,66 €
11	7,75 €	24	186,00 €
12	0,68 €	77	52,36 €
13	0,25 €	56	14,00 €
14	3,89 €	89	346,21 €
15	7,70 €	675	5.197,50 €
16	6,22 €	66	410,52 €
17	0,85 €	148	125,80 €
18	0,77 €	690	531,30 €
19	1,23 €	52	63,96 €
20	4,05 €	12	48,60 €

1. Identify SKUs where managers should spend time.
2. Prioritize SKUs according to their value to the company.
3. Create logical groupings
4. Adjust as necessary

Example:

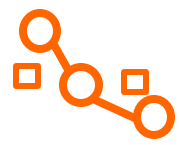
- Sample of 20 SKUs
- Total of 4,677 units
- Total ~\$22 thousand



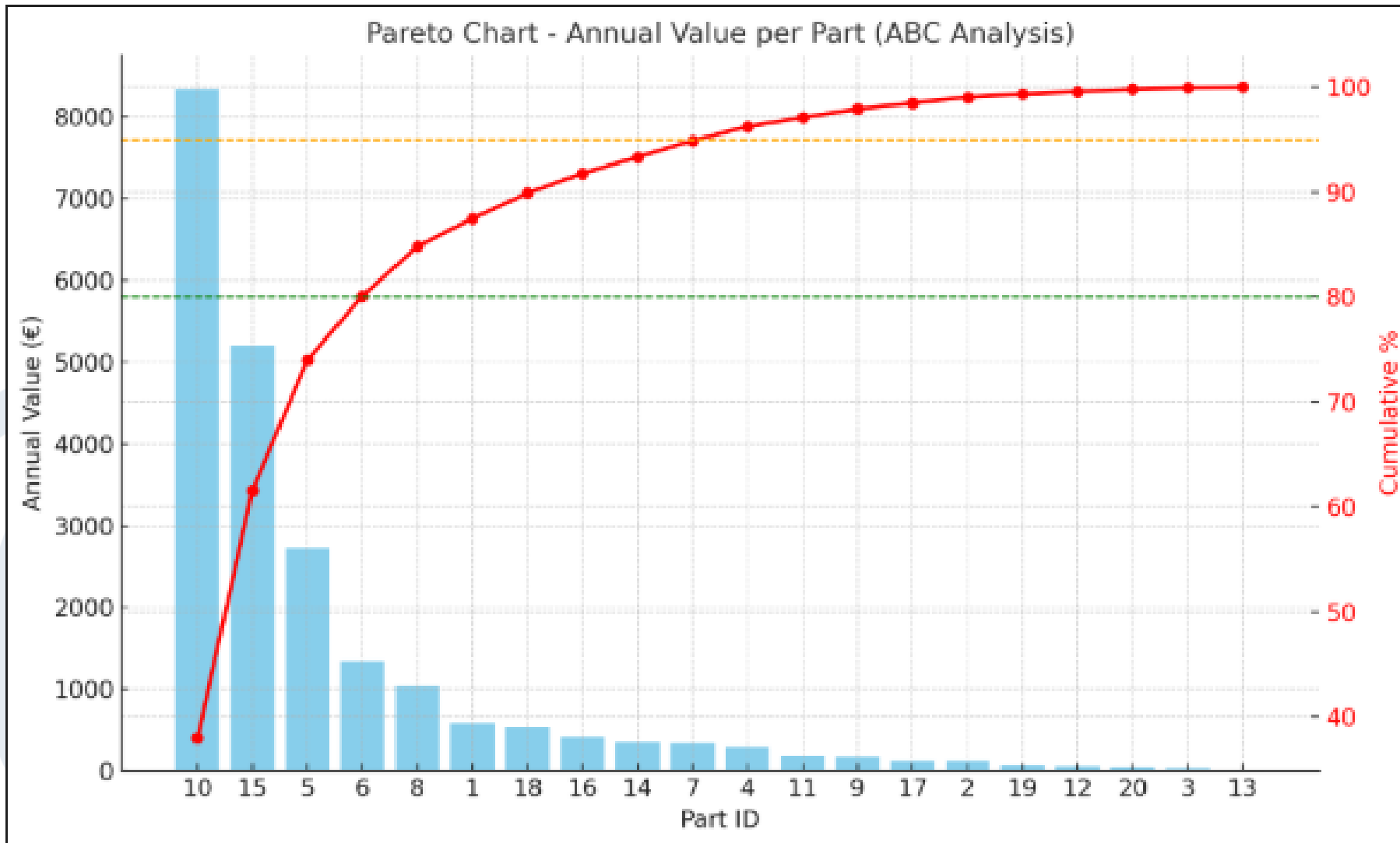
Segmentation: ABC Analysis

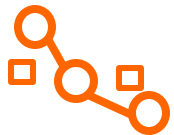
Part ID	Price	Annual Demand	Annual Value	Accumulate	% Annual	
10	24,99 €	334	8.346,66 €	8.346,66 €	38%	A
15	7,70 €	675	5.197,50 €	13.544,16 €	62%	
5	4,45 €	612	2.723,40 €	16.267,56 €	74%	
6	6,10 €	220	1.342,00 €	17.609,56 €	80%	
8	1,32 €	786	1.037,52 €	18.647,08 €	85%	
1	2,25 €	260	585,00 €	19.232,08 €	87%	B
18	0,77 €	690	531,30 €	19.763,38 €	90%	
16	6,22 €	66	410,52 €	20.173,90 €	92%	
14	3,89 €	89	346,21 €	20.520,11 €	93%	
7	3,10 €	110	341,00 €	20.861,11 €	95%	
4	0,77 €	388	298,76 €	21.159,87 €	96%	C
11	7,75 €	24	186,00 €	21.345,87 €	97%	
9	12,80 €	14	179,20 €	21.525,07 €	98%	
17	0,85 €	148	125,80 €	21.650,87 €	98%	
2	2,85 €	43	122,55 €	21.773,42 €	99%	
19	1,23 €	52	63,96 €	21.837,38 €	99%	
12	0,68 €	77	52,36 €	21.889,74 €	100%	
20	4,05 €	12	48,60 €	21.938,34 €	100%	
3	1,50 €	21	31,50 €	21.969,84 €	100%	
13	0,25 €	56	14,00 €	21.983,84 €	100%	
		4677	21.983,84 €			

< Important.

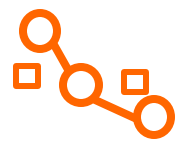


Segmentation: ABC Analysis



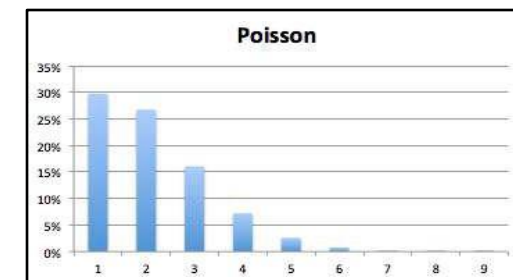
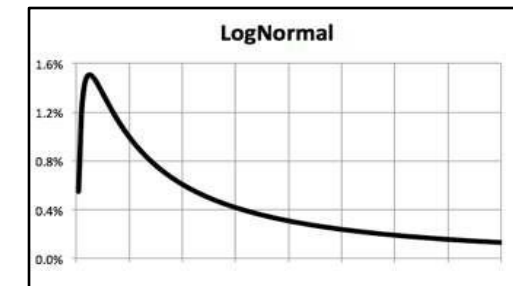
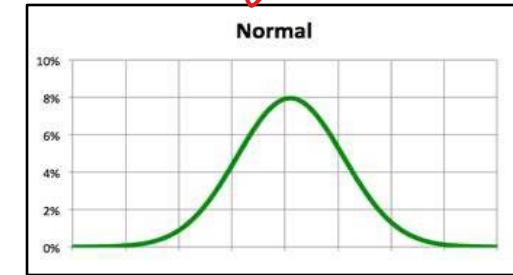


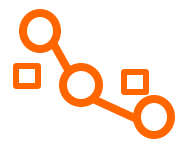
Uncertainty Management



Variability and Uncertainty

- It occurs in all aspects of supply chains.
- Using the "mean" or the "average" is rarely enough
- It is managed assuming a probability distribution
 - Normal distribution $\sim N(\mu, \sigma)$
 - Continuous ($-\infty < x < +\infty$) and Symmetric.
 - Most commonly used - for better or worse
 - Lognormal distribution $\sim \Lambda(\mu^*, \sigma^*)$
 - Continuous ($0 < x < +\infty$) and skewed to the right.
 - Less common, but good for supply chains
 - Poisson distribution $\sim P(\lambda)$
 - Discrete (integers ≥ 0)
 - Commonly used for low value distributions



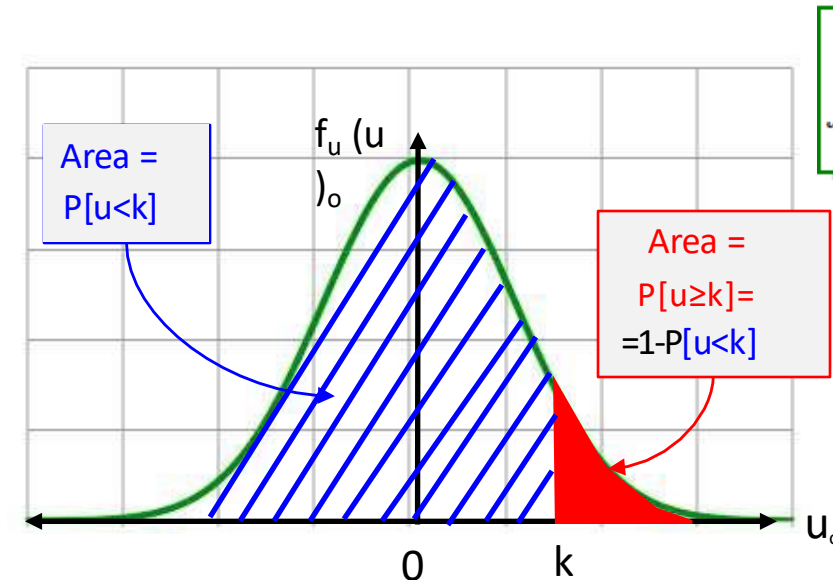
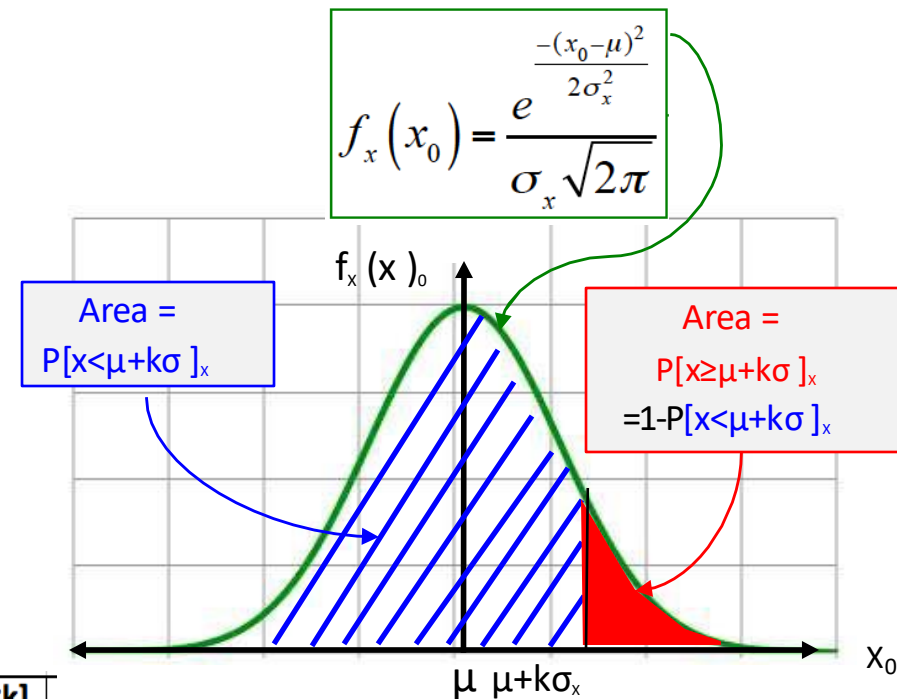


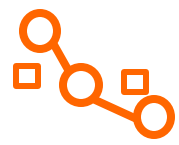
Normal distribution

- Normal $\sim N(\mu, \sigma)$
 - Spreadsheets
 - $\text{INV.NORM}(P[x_0 < x], \mu, \sigma) = x$
 - $\text{DISTR.NORM.N}(x, \mu, \sigma, 1) = P[x_0 < x]$
- Standard normal $\sim N(0, 1)$
 - Transformation: $k = (x - \mu) / \sigma_x$
 - Spreadsheets
 - $\text{INV.NORM.ESTAND}(\text{probability}) = k$
 - $\text{DISTR.NORM.ESTAND}(k) = P[u < k]$
- Normal tables of standard units
 - Search k or $P[u < k]$

$\mu \pm \sigma$	68,3%
$\mu \pm 2\sigma$	95,5%
$\mu \pm 3\sigma$	99,7%

k	P[u < k]
0.50	0.6915
0.51	0.6950
0.52	0.6985
0.53	0.7019
0.54	0.7054
0.55	0.7088
0.56	0.7123





Poisson distribution

- Poisson $\sim P(\lambda)$

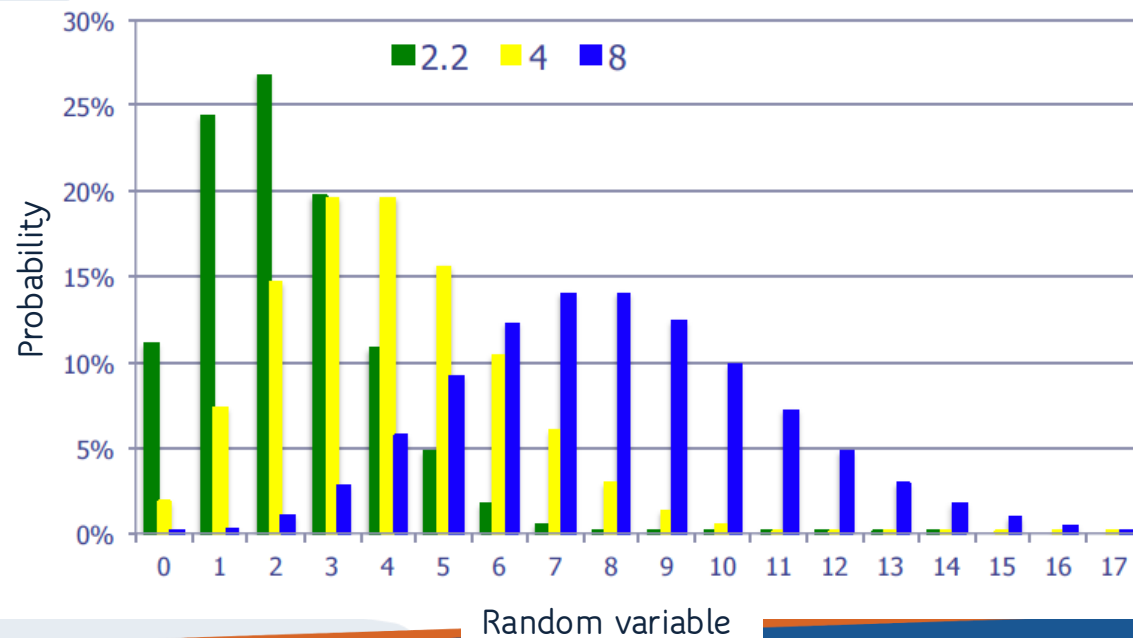
- Probability of occurrence of x events over a period of time
- Mean = Variance = λ

$$p[x_0] = \text{Prob}[x = x_0] = \frac{e^{-\lambda} \lambda^{x_0}}{x_0!} \quad \text{for } x_0 = 0, 1, 2, \dots$$

$$F[x_0] = \text{Prob}[x \leq x_0] = \sum_{x=0}^{x_0} \frac{e^{-\lambda} \lambda^x}{x!}$$

- In Spreadsheets:

- $p(x_0) = \text{POISSON}(x_0, \lambda, 0)$
- $F(x_0) = \text{POISSON}(x_0, \lambda, 1)$



Poisson's tables (partial)

- Columns: λ
- Rows: $F(x)_0$

F(x)	0.75	1.00	
0	0.47237	0.36788	(
1	0.82664	0.73576	(
2	0.95949	0.91970	(
3	0.99271	0.98101	(
4	0.99894	0.99634	(
5	0.99987	0.99941	(
6	0.99999	0.99992	(
7	1.00000	0.99999	(